METRIC

DOD-P-15817D(SH)
18 August 1976
SUPERSEDING
MIL-P-15817C(SHIPS)
13 November 1956
(See 6.3)

MILITARY SPECIFICATION

PARTS, SYNTHETIC RUBBER, FOR USE WITH BATTERIES OR IN BATTERY COMPARTMENTS OF SUBMARINES (METRIC)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers synthetic rubber parts for use in batteries or battery compartments of submarines.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-676 - Boxes, Setup.

MILITARY
MIL-P-116 - Preservation-Packaging, Methods of.
MIL-S-8660 - Silicone Compound.

STANDARDS

FEDERAL
FED-STD-601 - Rubber: Sampling and Testing.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

FSC 9320

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UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois, 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The synthetic rubber used shall be free from metallic particles and from lumps or granular indications that the aggregates have not been thoroughly mixed into the compound. Surfaces shall present a neat, smooth, and finished appearance, and shall be free from pitting, fissures, blowholes, rough spots, rubber scale, blisters, and other deformations.

3.2 Form, dimensions, and tolerances. Form, dimensions, and tolerances shall be in conformance with the drawings provided for the parts specified (see 6.2).

3.3 Tensile strength.

3.3.1 Before aging. Before aging, tensile strength shall be not less than 8.3 megapascals (MPa) when tested as specified in 4.3.3.

3.3.2 After aging. After completion of the aging process specified in 4.3.2, the tensile strength shall be not less than 75 percent of the tensile strength of unaged samples.

3.4 Ultimate elongation.

3.4.1 Before aging. Before aging, ultimate elongation shall be not less than 300 percent when tested as specified in 4.3.4.

3.4.2 After aging. After completion of the aging process of 4.3.2, ultimate elongation shall be not less than 70 percent of the percent ultimate elongation of unaged samples.

3.5 Hardness.

3.5.1 Before aging. Before aging, hardness shall be Shore A durometer hardness of between 43 and 52 when tested as specified in 4.3.5.

3.5.2 After aging. After completion of the aging process specified in 4.3.2, hardness shall be Shore A durometer hardness of 57 maximum when tested as specified in 4.3.5.

3.6 Tension set. Tension set shall be not greater than 8.0 percent when tested as specified in 4.3.6.

3.7 Compression set. This requirement shall apply to material intended for gaskets and washers.

3.7.1 Before aging. Before aging, compression set shall be not greater than 50 percent when tested as specified in 4.3.7.

3.7.2 After aging. After completion of the aging process specified in 4.3.2, compression set shall be not greater than 40 percent when tested as specified in 4.3.7.

3.8 Resistance to acid and alkali. Volume swell shall be not greater than 8.0 percent when tested as specified in 4.3.8.

3.9 Sealing ability. This requirement shall apply to material intended to be used for ventilation duct sleeves. When specimens are tested as specified in 4.3.9, there shall be no leakage.

3.10 Environmental requirements. Rubber parts shall not be adversely affected by storage in environmental temperatures of minus 30°C to 55°C or minus 15°C to 55°C when installed for use.

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3.11 Marking. Molded articles shall have the piece number or other identifying marking molded as raised or impressed characters on the surface. Articles manufactured from sheets and tubes shall be marked with the piece number in an ineradicable manner, as by stamping or branding. If marking would impair the usefulness of the part, nonpermanent marking as by stencil or tag may be used (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system in accordance with the data ordering document included in the contract or order (see 6.2.1).

4.2 Quality conformance inspection.

4.2.1 Lot.

4.2.1.1 Visual and dimensional examination. Parts of the same type made in a production run shall be considered a lot for purposes of visual and dimensional examination.

4.2.1.2 Tests. A lot shall consist of material produced in one process batch, having the same application, and offered for delivery at one time. Lot size shall be equal to the mass of the lot in kilograms (kg).

4.2.2 Sampling.

4.2.2.1 Visual and dimensional examination. A random sample of sleeves or parts shall be selected from each lot in accordance with MIL-STD-105. The Acceptable Quality Level (AQL) shall be 1.5 percent defective. The samples shall be examined as specified in 4.3.1.

4.2.2.2 Tests. A random sample of sleeves or parts for each test shall be selected from each lot in accordance with MIL-STD-414 at inspection level I and shall be subjected to the tests of 4.3. The AQL shall be 1.5 percent defective. When the specified determinations are not possible from samples of the material comprising the lot, the contractor shall furnish the number of sample pieces required by MIL-STD-414 for the test specified in 4.3. These sample pieces shall be identical in composition and equivalent in cure, and prepared at the same time from material used in the lot of sleeves or parts offered for delivery.

4.3 Test methods.

4.3.1 Visual and dimensional examination. Each of the samples selected in accordance with 4.2.2.1 shall be visually and dimensionally examined to verify compliance with 3.1, 3.2, 3.10, and 3.11 using criteria of MIL-STD-407.

4.3.2 Aging process. Specimens shall be subjected to an aging process by exposure in an oven to dry circulating air at a temperature of 90°C ± 1°C for no less than 46 hours + 1/4 hour. Tests on specimens before aging shall be performed immediately prior to and on the same machine as corresponding tests performed on specimens after aging. The latter tests shall be performed no less than 20 hours and no more than 48 hours following completion of the aging process.

4.3.3 Tensile strength. Tensile strength tests and specimens used shall comply with method 4111 and with die II of the dumbbell test specimen of FED-STD-601 except as otherwise required by excessive elongation (4.3.4).

4.3.4 Ultimate elongation. Method 4121 and die II test specimens for ultimate elongation of FED-STD-601 shall be used. Die III of the dumbbell test specimen may be used when the elongation of the synthetic rubber is such that the specimen cannot be broken within the limits of the tensile testing machine.

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4.3.5 Hardness. Hardness shall be determined using method 3021 of FED-STD-601.

4.3.6 Tension set. Tension set shall be determined by method 4411 of FED-STD-601.

4.3.7 Compression set. Compression set shall be determined by method 3311 of FED-STD-601.

4.3.8 Resistance to acid and alkali. Specimens shall be immersed in a solution of 60 percent (by mass) sulfuric acid and in a solution of 45 percent (by mass) potassium hydroxide using approximately 50 milliliters (ml) of solution per specimen. Specimens shall remain fully submerged for 168 + 1 hours at 70°C + 1°C and shall be measured for volume swell in accordance with method 11421 of FED-STD-601.

4.3.9 Sealing ability. A test jig shall be used which is machined from a cylindrical block of hard rubber to the shape shown on figure 1 and to the dimensions shown on figure 2. The diameters of the lands of each jig shall be equal, plus or minus 0.1 millimeters (mm), to the nominal outside diameter of the hard rubber ducts for which the sleeves under test are intended. The land surfaces of the jigs shall have a roughness of not more than 150 micrometers (µm). The test specimen shall be a section 32 mm long cut from a sleeve. The land surfaces of the jig and the inside of the specimen shall be smeared with silicone grease in accordance with MIL-S-8660. The specimen shall t be positioned on the jig as shown on figure 3, and the jig filled with sulfuric acid so on 60 percent (by mass) to a height of 50 mm above the test groove, or 35 mm above the top edge of the test specimen. A stopper and glass tube shall be inserted as shown on figure 3 to contain the acid at the desired level. The glass tube shall be plugged with glass wool to reduce evaporation. The assembly shall be placed in a circulating air oven for 168 + 1 hours (7 days) at 70°C + 1°C. If no leakage has been observed by the end of this period, the assembly shall be taken apart, put back together again, refilled with acid, and conditioned for an additional 25 + 1/4 hours at 70°C + 1°C. Additional sealing compound shall be supplied to the sleeve and jig if necessary. Leakage shall be determined by observation. The test above shall be repeated using a solution of 45 percent (by mass) potassium hydroxide using petrolatum or any other petroleum based grease in place of silicone grease.

4.4 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Parts shall be cleaned by process C-1 of MIL-P-116. Solvents may be used provided they do not contain petroleum or aromatic hydrocarbons or other material which may damage or cause degradation of the products.

5.1.1.2 Drying. Parts shall be dried in accordance with one or more of the drying procedures listed in MIL-P-116.

5.1.1.3 Unit packaging. Synthetic rubber parts shall be individually packaged in accordance with method III of MIL-P-116.

5.1.1.4 Intermediate packaging. When 10, or less, identical items are being shipped at the same time to one destination, intermediate packaging will not be required. The item shall be packaged as specified herein and packed directly into shipping containers as specified for level A, B, or C, as applicable. When more than 10 identical items are being shipped at the same time to one destination, the items shall be intermediate packaged in quantities of not more than 10 in an intermediate package. Intermediate containers shall conform to PPP-B-566, PPP-B-676, or PPP-B-636, at the option of the contractor. Box closures shall conform to the applicable box specification and the appendix thereto, except that when PPP-B-636, class weather-resistant boxes are used, method V closure shall be applied. The gross mass of paperboard boxes shall not exceed 4.3 kg and the gross mass of fiberboard boxes shall not exceed 9.0 kg.

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5.1.2 Level C. Cleaning, drying, preservation-packaging of rubber parts shall afford protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. The contractor's normal retail or wholesale preservation-packaging methods may be utilized when such meet the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Parts, packaged as specified in 5.1, shall be packed in fiberboard boxes conforming to PPP-B-640, class 2, or PPP-B-636, class weather-resistant, grade V. Box selection, variety, and style shall be at the contractor's option. Boxes shall be closed, reinforced, and water-proofed by the use of nonmetallic tape banding. For PPP-B-636 boxes, method V closure in accordance with the appendix to box specification shall be applied. The gross mass of triple wall boxes shall not exceed 90 kg; fiberboard boxes shall not exceed the mass limitation of the box specification.

5.2.2 Level B. Parts, packaged as specified in 5.1, shall be packed in fiberboard boxes conforming to PPP-B-640, class 1, or PPP-B-636, class-domestic, minimum 350 grade. Box selection, variety, and style shall be at the contractor's option. Closure shall be specified in the appendix to the applicable box specification with method I closure applying to PPP-B-636 boxes. Intermediate fiberboard boxes closed as specified herein and used as shipping containers need not be overpacked. The gross mass of triple wall boxes shall not exceed 90 kg; fiberboard boxes shall not exceed the mass limitation of the applicable box specification.

5.2.3 Level C. Parts, preserved-packaged as specified in 5.1, shall be packed in containers acceptable to the common carrier which will ensure safe delivery at the destination in a satisfactory condition at the lowest applicable rate. Containers or method of packing shall comply with Uniform Freight or National Motor Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking specified in the contract or order, unit package and intermediate package and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Synthetic rubber parts covered by this specification are intended for use in propulsion batteries, battery parts, battery repair, or maintenance parts and in battery compartments of submarines.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable drawing, showing form of material, dimensions, and tolerances required (see 3.2), and marking required (see 3.11).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Contract data requirements (see 6.2.1).

6.2.1 Contract data requirements. When this specification is used in a procurement invoking the provisions of Armed Service Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the following data requirements should be specified for delivery on the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the following data should be specified for delivery in accordance with the contract or order.

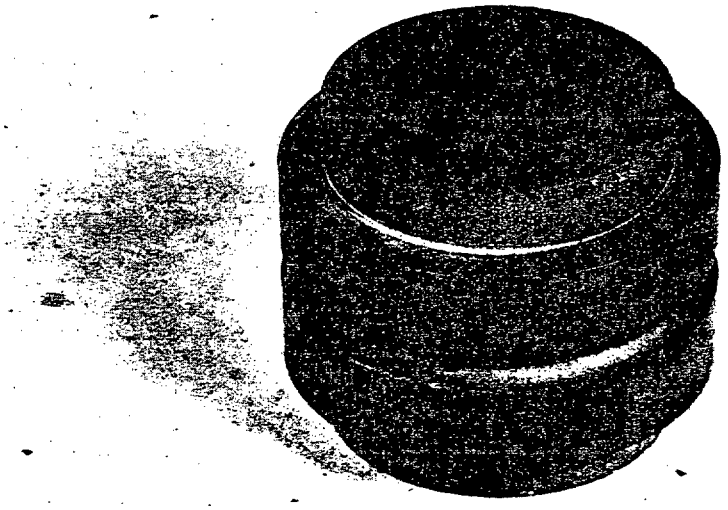
Specification paragraph	Data requirement	Service	Applicable DID	Options
4.1.1	Inspection System Program Plan	SH	DI-R-4803	-----

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 9320-N211)

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SH 1795

FIGURE 1. Jig for testing battery ventilation duct sleeve sections.

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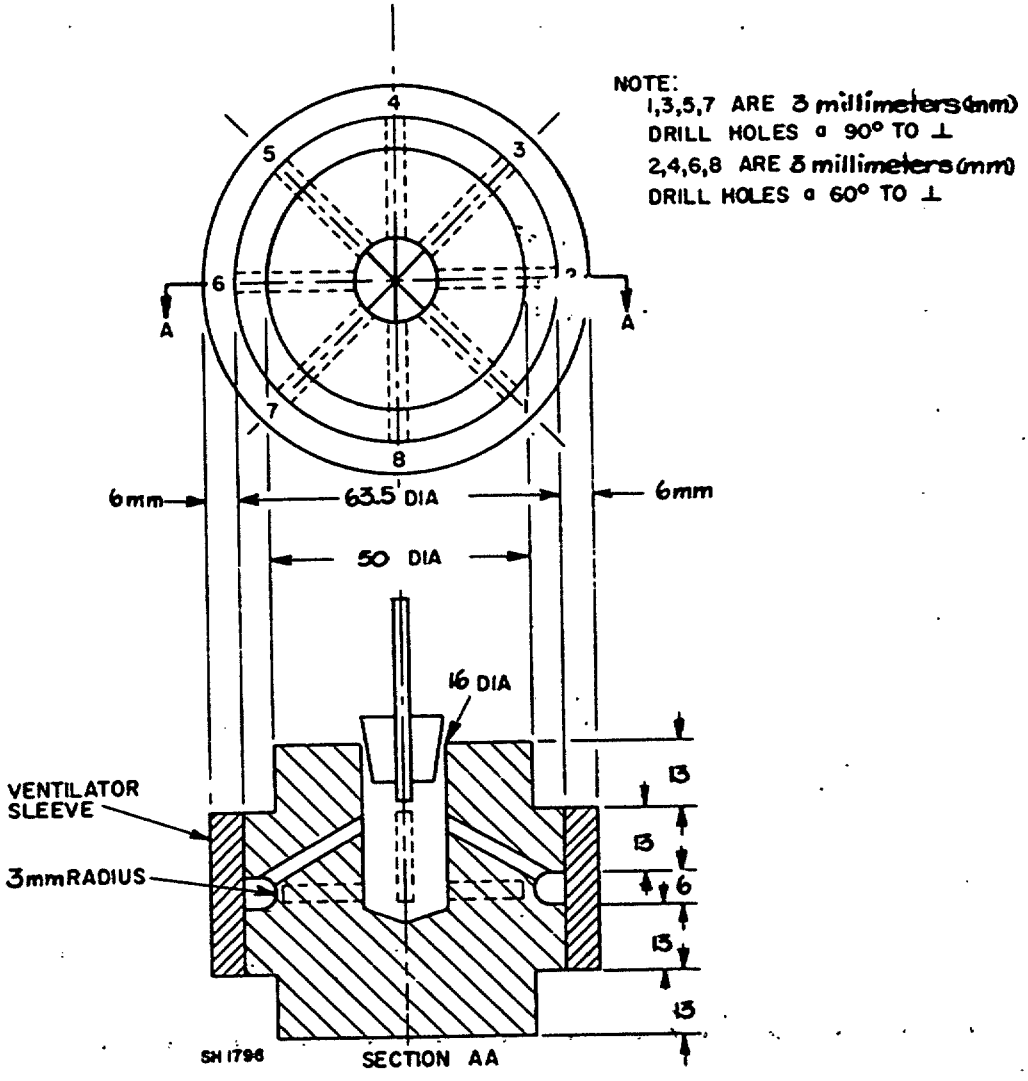
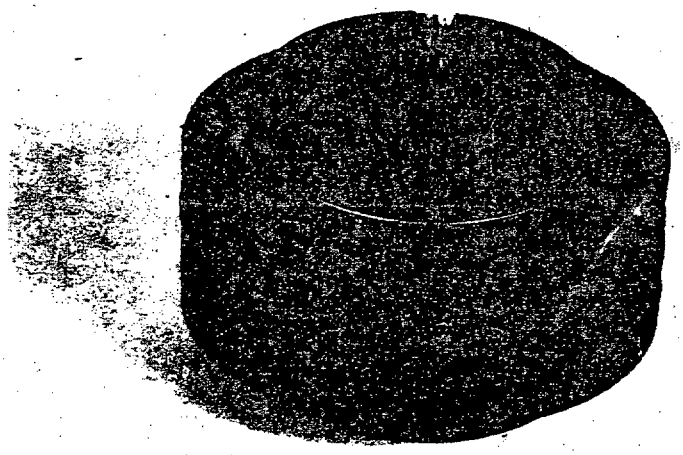


FIGURE 2. Jig for testing battery ventilation duct sleeve sections.
(All dimensions in millimeters)

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SH 1787

FIGURE 3. Battery ventilation duct sleeve section in place on test jig.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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